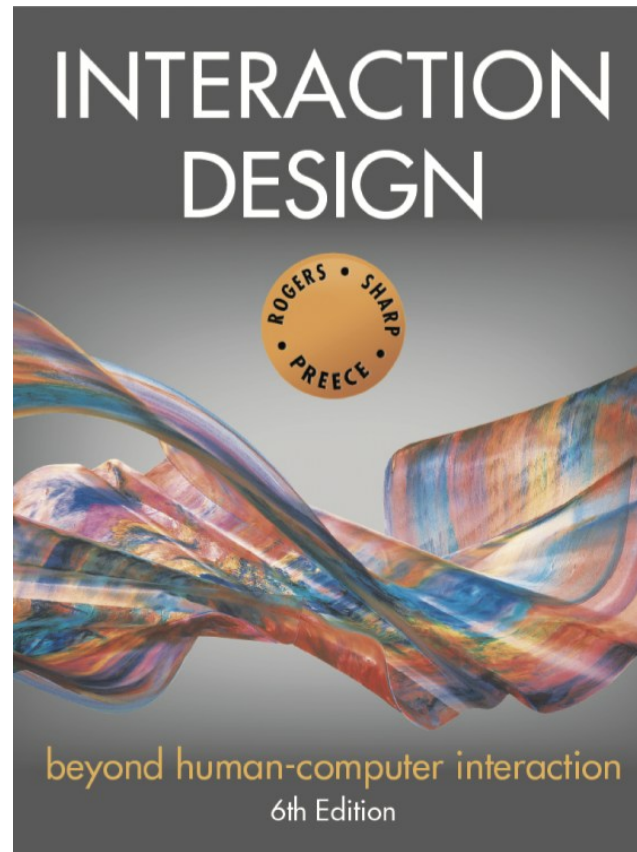




Chapter 3

CONCEPTUALIZING INTERACTION

Learning Goals

- Explain **how to conceptualize interaction**.
- Describe what a **conceptual model** is and how to begin to formulate one.
- Discuss the use of **interface metaphors** as part of a conceptual model.
- Outline the **core interaction types** for informing the development of a conceptual model.
- Introduce paradigms, visions, challenges, theories, models, and frameworks informing interaction design.

Conceptualizing design

Conceptualize **what the proposed product will do**

Why the need to conceptualizing design?

- To scrutinize vague ideas and assumptions about the benefits of the proposed product and user experience in terms of their feasibility
- How realistic is it to develop?
- How desirable and useful?
- Many unknowns need to be considered in the initial stages of a design project, especially if it is a new product that is being proposed
 - Conceptualization helps to reduce the risks

Consider design example 1

- A designer comes up with the idea of creating a **speaking mobile robot waiter**:
 - help waiters in a restaurant take orders and deliver meals to customers
- Why ? What problem would this address?
- Putative (or supposed) benefits
 - Could help take orders and entertain customers by having a conversation with them at the table
- **Design concerns**
 - How would it need to move to appear to be talking?
 - What would the customers think of it?
 - Would it always be a pleasure for them to engage with the robot, not knowing what it would say on each new visit to the restaurant?
 - Would it be annoying or a fad?

Consider design example 2

- An **automated robot server** that is functional
- In contrast to the idea of a **speaking robot waiter**, the (non-speaking) **robot server** addressed a real-world problem
 - the cooks load up the trays on its shelves, and then sends it to a preset area of the dining room to deliver the food
 - the customers remove the food and drinks from the robot server
- It is like a tool, there to save waiters' time rather than replacing them

Which type of server is preferable and on what grounds?



A humanoid **nonspeaking robot server** in Shanghai. Imagine what would be gained if it could also have a conversation with customers.

Which type of server is preferable and on what grounds?

A non-humanoid **robot server** with a set of trays that orders are placed onto



Assumptions and claims

- Write down your assumptions and claims when deciding which one you prefer
- Try to defend and support them by what the two different servers will provide

What is an assumption?

- Taking something for granted when it needs further investigation, e.g.
 - The robot could take orders and entertain customers by having a conversation with them
 - The robot could make recommendations for different customers, such as restless children or fussy eaters

What is a claim?

- A claim is stating something to be true when it is still open to question
 - It will raise revenue by attracting more people to the restaurant
 - It will solve the labor crisis of there being not enough human waiters
- The real problem being addressed:

“It is difficult to recruit good wait staff who provide the level of customer service to which we have become accustomed.”

Working through assumptions

- Need to work through the initial stages of a design project
 - Where do your ideas come from?
 - What sources of inspiration were used?
 - Is there any theory or research that can be used to inform (support) them?
- During the early ideation process
 - Ask questions, reconsider assumptions, and articulate concerns

Poor assumptions

- Those that are difficult to articulate
 - can highlight what ideas are vague or unrealistic
 - identify human activities and interactivities that are problematic
- Based on this kind of conceptual analysis iteratively work out how the design ideas might be improved

A framework for analyzing the problem space

Understanding the Problem: Ask critical questions:

- Are there problems with an existing product or user experience? If so, what are they?
- Why do you think there are problems?
- How do you think your proposed design ideas might overcome these?
- If you are designing for a new user experience, how do you think your proposed design ideas support, change, or extend current ways of doing things?

Activity

- What were the assumptions and claims made about watching 3D TV?



Figure 3.2 A family watching 3D TV

Source: Andrey Popov, [Shutterstock](#)

Assumptions and claims: how realistic?

- There was no existing problem to overcome
 - What was being proposed was a new way of experiencing TV
- An assumption
 - People would really enjoy the enhanced clarity and color detail provided by 3D
- A claim
 - People would not mind paying a lot more for a new 3D-enabled TV screen because of the new experience

Benefits of conceptualizing

Orientation

- Enables design teams to **ask specific questions** about how the **conceptual model** will be understood

Open-minded

- Prevents design teams from becoming narrowly focused early on

Common ground

- Allows design teams to establish a set of commonly agreed terms

From problem space to design space

- Having a good understanding of the problem space can help inform the design space
 - for example, what kind of interface, behavior, functionality to provide
- Before deciding upon these, it is important to develop a conceptual model

Conceptual model

- A conceptual model is:
“...a high-level description of how a system is organized and operates” (Johnson and Henderson, 2002, p26)
- It enables:
“...designers to straighten out their thinking before they start laying out their widgets” (Johnson and Henderson, 2002, p28)
- Provides a working strategy and framework of general concepts and their interrelations

Conceptual Model

- Designer's (i.e., your) mental model
 - What users will be able to do
 - What concepts or knowledge users will need, in order to interact
 - How they will interact with system (at a very high level)

Components

- Metaphors and analogies
 - A **metaphor**, as you may know, is a figure of speech that **refers to one thing by mentioning another**.
 - For example, a "desktop" in a computer system is a metaphor for a physical desk where files and documents are organized.
 - Help understand what a product is for and how to use it for an activity

Components (Cont.)

- Concepts that people are exposed to through the product
 - Task–Domain **objects**(the things users interact with), their **attributes** (characteristics of these objects), and **operations** (actions users can take on the objects).
 - For example, in a file management system,
 - the task-domain objects might be "files" or "folders,"
 - their attributes could include "name" or "size," and
 - operations could involve "saving," "revisiting," or "organizing."
- Relationship and mappings between these concepts

First steps in formulating a conceptual model

- What will people be doing when carrying out their tasks?
- How will the proposed model support these?
- What kind of interface metaphor, if any, will be appropriate?
- What kinds of interaction modes and styles to use?
- Keep in mind when making design decisions how the user will understand the underlying conceptual model

Conceptual models

- Many kinds and ways of classifying them
- The best conceptual models are often those that appear:
 - Obvious and simple
 - The operations they support are intuitive to use

Metaphors

- A **metaphor** is a **figure of speech** that describes an object or action by referring to another object or action, suggesting a similarity between them,
 - e.g., world is a stage, He is a movie star.
- Well known concepts you can rely on to help users understand and interact with the system
- **Interface metaphors** help users to understand the functionality and content of a system quickly and intuitively.

Interface metaphors

- Interface designed to be similar to a physical entity but also has own properties
 - e.g, desktop metaphor, web portals, search engine
- Can be based on activity, object, or a combination of both
- Exploit user's familiar knowledge, helping them to understand 'the unfamiliar'
- Conjures up the essence of the unfamiliar activity, enabling users to leverage this to understand more aspects of the unfamiliar functionality

Examples of interface metaphors

- Conceptualizing what users are doing (**an activity**)
 - e.g, **Surfing the Web**: Web surfing is similar to surfing the ocean. It conveys the idea of navigating through a vast ocean of information.
- A conceptual model instantiated at the **user interface**
 - e.g., **the desktop metaphor**
- Visualizing **an operation**
 - e.g., an icon of a **shopping cart** into which the user places items

A Classic Conceptual Model: The Xerox Star

- The Star interface, developed by Xerox in 1981
- it was based on a conceptual model that included the familiar knowledge of an office.

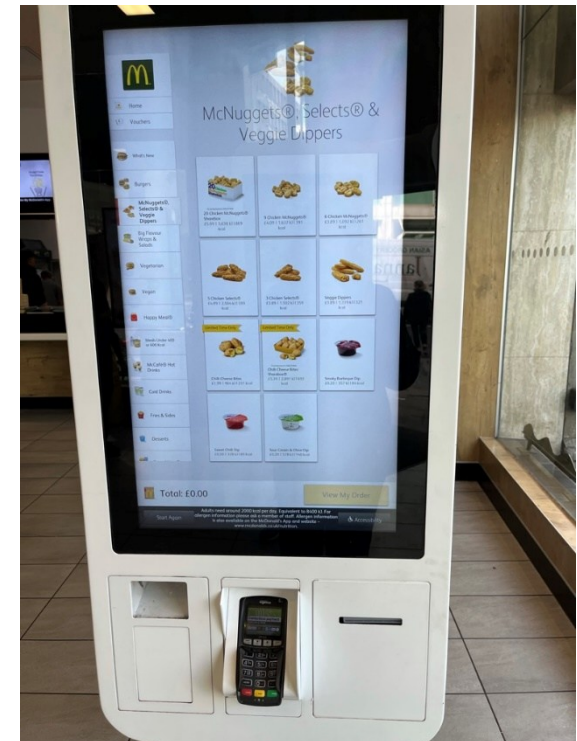


A Classic Conceptual Model: The Xerox Star

- Paper, folders, filing cabinets, and mailboxes were represented as icons on the screen.
- Dragging a document icon across the desktop screen was seen as equivalent to picking up a piece of paper in the physical world and moving it
- Dragging a digital document into a digital folder was seen as being analogous to placing a physical document into a physical cabinet

The card metaphor

- A **self-ordering kiosk** found in many fast-food restaurants that uses a **card metaphor** as part of the interface.
- Each food or drink item is displayed on its own card with associated description and price.
- This enables the customer to **easily select items** they want to purchase from the category types in the card matrix while also going back and forth between the other high-level categories (e.g., burgers, vegan, cold drinks), shown down the left side of the display.



A fast food display

The card metaphor

- The **card** is a very popular **metaphor**.

Why?

- It has familiar form factor
- It can easily be flicked through(scrolled), sorted, and themed(categorized)
- Cards have strong associations in the user's mind, providing an intuitive way of organizing content that is “card sized”
- Structures content into meaningful chunks

Benefits of interface metaphors

- Makes learning new systems easier
- Familiar artefact or action helps people understand the underlying conceptual model
- Can be innovative and enable apps to be made accessible to a greater diversity of people

Activity

- Describe the components of the conceptual model underlying most online shopping websites, for example:
 - Shopping cart
 - Proceeding to check-out
 - 1-click
 - Gift wrapping
 - Cash register

'Shopping Cart' Metaphor

- The concept of a '**shopping cart**' on e-commerce websites is a metaphor.
- In real life, you grab a cart, fill it with items, and then proceed to check out.
- Translating this concept to the online platform, users easily understand what a 'shopping cart' is meant to do.
- The **real-life process of shopping** is metaphorically applied to the website, thereby making the user interface simple and familiar.

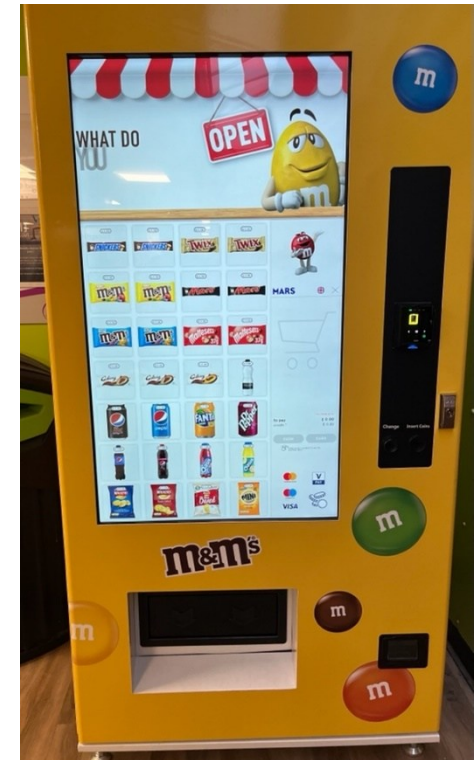
Interaction types

- Instructing
 - Issuing commands and selecting options
- Conversing
 - Interacting with a system as if having a conversation
- Manipulating
 - Interacting with objects in a virtual or physical space by manipulating them
- Exploring
 - Moving through a virtual environment or a physical space
- Responding
 - The system initiates the interaction and the person chooses whether to respond

1. Instructing

- Where someone instruct a system and tells it what to do
 - e.g. Tell the time, print a file, or save a file
- Very common conceptual model underlying a diversity of devices and systems
 - e.g. Word processors, VCRs or DVD players, vending machines
- The main benefit is that instructing supports quick and efficient interaction
 - Good for repetitive kinds of actions performed on multiple objects

Which is easiest and why?



First Vending Machine

- The first vending machine has been designed using **simple instructions**.
- There is a small number of drinks to choose from and each is represented by a **large button displaying the label of each drink**.
- The user simply has to **press one button** and this should have the effect of returning the selected drink.

Second Vending Machine

- The second machine is more complex, offering a wider range of snacks
- Uses a more complex process, involving:
 - (i) reading off the code (for example, C12) under the item chosen, then
 - (ii) keying this into the number pad adjacent to the displayed items, and
 - (iii) checking the price of the selected option and ensuring that the amount of money inserted is the same or greater
- Problems:
 - customer misreading the code and/or miskeying in the code

Third Vending Machine

- The third vending machine has a **large digital touch screen** on the front.
- It provides the best of both worlds by providing a **direct manipulation-style interface**.
- The images that appear on the **display clearly show what the options are** and beside each **their price**.
- The customer needs only to **press the display of the object** chosen and place their credit/debit phone on the card reader on the right.
- Much lower chance of error resulting from pressing the wrong code or keys.

2. Conversing

- Underlying model of having a conversation with another human
- Ranges from **simple voice recognition** menu-driven systems (for example, A phone menu where you say "Press 1 for customer service" or "Press 2 for billing.") to more complex **'natural language' dialogs** (For example, A chatbot or virtual assistant (like Siri or Alexa) that responds to complex, open-ended questions such as "What's the weather like today?" or "Set a reminder for 3 PM.")

Conversing (Cont.)

- Examples include
- Timetables: What time does the next train leave?
- search engines: How tall is the Eiffel Tower?
- advice-giving systems: medical/financial advice, etc.
- help systems: How do I reset my password?
- virtual agents, chatbots, toys, and pet robots designed to converse with people

Pros and cons of conversational model ⁽¹⁾

- Allows people to interact with a system in a way that is familiar to them
 - can make them feel comfortable, at ease, and less scared
- Misunderstandings can arise when the system does not know how to parse what someone says
 - e.g, voice assistants can misunderstand what children say



Siri's response when asked if a person needs an umbrella today

Pros and cons of conversational model ⁽²⁾

- **Problems:**
 - Certain kinds of tasks are transformed into cumbersome and one-sided interactions.
 - This is especially true for **automated phone-based systems** that use **auditory menus** to advance the interaction.
 - Users have to listen to a voice providing several options, then make a selection, and finally repeat through further layers of menus before accomplishing their goal, for example, reaching a real human or paying a bill.

Pros and cons of conversational model ⁽²⁾

Dialog between a user and an insurance company

<user dials an insurance company>

“Welcome to St. Paul’s Insurance Company. Press 1 if you are a new customer; 2 if you are an existing customer.”

<user presses 1>

“Thank you for calling St. Paul’s Insurance Company. If you require house insurance, say 1; car insurance, say 2; travel insurance, say 3; health insurance, say 4; other, say 5.”

<user says 2>

“You have reached the car insurance division. If you require information about fully comprehensive insurance, say 1; third-party insurance, say 2. . . .”

Pros and cons of conversational model ⁽³⁾



**“If you’d like to press 1, press 3.
If you’d like to press 3, press 8.
If you’d like to press 8, press 5...”**

3. Manipulating

- Direct action on digital/virtual objects
- Involves dragging, selecting, opening, closing and zooming(in/out), stretching, shrinking actions on virtual objects
- Exploit's someone's knowledge of how they move and manipulate in the physical world
- Can involve actions using physical remote controllers or air gestures to control the movements of an on-screen avatar
- Physical objects (like balls, bricks, or toys) can be equipped with **sensors or tags** to **connect actions in the physical world with digital reactions.**

Direct Manipulation

- **Digital objects** in the interface can be designed so that they **can be interacted with in ways that are analogous** to how **physical objects** in the physical world are manipulated.
- Direct manipulation interfaces are assumed to **enable users to feel that they are directly controlling the digital objects** represented by the computer

Direct Manipulation

- Ben Shneiderman (1983) coined the term
- **Three core properties:**
 - Continuous representation of objects and actions of interest
 - Physical actions and button pressing instead of issuing commands with complex syntax
 - Rapid, reversible actions with immediate feedback on object of interest
- According to these principles:
 - **an object on a screen remains visible** while a user performs physical actions on it, and **any actions performed on it are immediately visible**

Benefits of direct manipulation

- Novices can learn the basic functionality quickly
- Experienced users can carry out a wide range of tasks efficiently and rapidly
- Intermittent users can retain operational concepts over time
- Error messages rarely needed
- Users can immediately see if their actions are furthering their goals, and if not, do something else
- Users experience less anxiety
- Users gain confidence and mastery and feel in control

4. Exploring

- Involves moving through virtual or physical environments
 - People can explore aspects of a 3D environment
 - Physical environments can also be embedded with sensors that when detect the presence of someone will trigger digital or physical events to happen
- Many examples of virtual environments, including cities, parks, buildings, rooms, and datasets
 - enable people to fly over them and zoom in and out of different parts

Seeing things larger than life and in situ in AR



A virtual lion appearing in someone's living room created with Google 3D object

Exploring data in VR



Someone using Immersive data analytics allowing them to explore Tableau data in 3D

Responding*

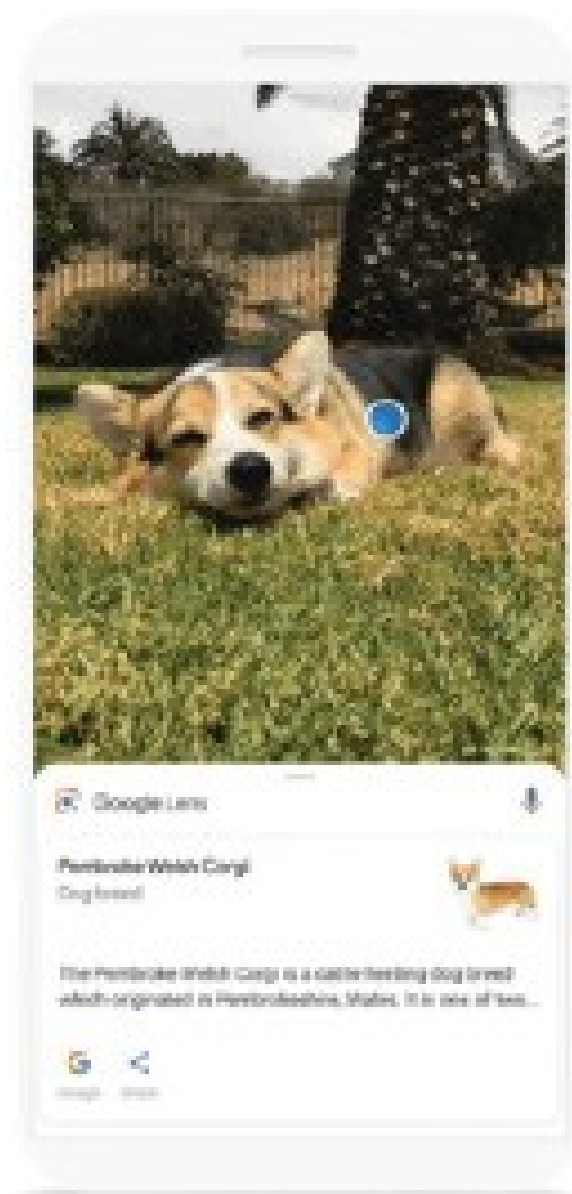
- System takes the initiative to alert user to something that it “thinks” is of interest
- System does this by:
 - detecting the location and-or presence of someone in a vicinity and notifying them on their phone or smartwatch
 - what it has learned from their repeated behaviors
- Examples:
 - alerts the user of a nearby coffee bar where some friends are meeting
 - A wearer’s fitness tracker notifies them of a milestone reached
- Automatic system response without any requests made by the user

* This type suggested by Christopher Lueg et al. (2018)

Example:

After taking a photo of a friend's cute dog in the park, Google Lens will automatically pop up information that identifies the breed of the dog

Figure - Google Lens in action, providing pop-up information about Pembroke Welsh Corgi having recognized the image as one.



Potential cons of system-initiated notifications

- Can get tiresome or frustrating if too many notifications or the system gets it wrong
- What does it do when it gets something wrong?
 - Does it apologize?
 - Does it allow the user to correct the advice or information?

Choosing an interaction type

- Direct manipulation is good for 'doing' types of tasks, for example, designing, drawing, flying, driving, or sizing windows
- Issuing instructions is good for repetitive tasks, for example, spell-checking and file management
- Having a conversation is good for finding out information, requesting music/TV or learning
- Hybrid conceptual models are good for supporting multiple ways of carrying out the same actions

Difference between interaction types and interface styles

Interaction type:

- A description of what a person is doing when interacting with a system, for example, instructing, talking, browsing, or responding

Interface style:

- The kind of interface used to support the interaction, for instance, command, menu-based, gesture, or voice

Many kinds of interface styles available (see Chapter 7)

- Command
- Speech
- Data-entry
- Form fill-in
- Query
- Graphical
- Web
- Pen
- Augmented reality
- Gesture

Summary

- Developing a conceptual model involves:
 - Understanding the problem space
 - Being clear about the underlying assumptions and claims
 - Specifying how the proposed design will support people in their activities
- A conceptual model is a high-level description of a product in terms of:
 - What users can do with it and the concepts they need to understand how to interact with it
- Interaction types provide a way of thinking about how to support user's activities